Abdullah Farhat

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Education

B.E. Computer Engineering (dual degree), **Dartmouth College** Hanover, NH | June 2029

Coursework in progress: Distributed Systems and Fields, Embedded Systems, Digital Electronics.

B.A. Computer Science, Minor in Mathematics, **Vassar College** Poughkeepsie, NY | May 2028

Coursework: Data Structures & Algorithms, Software Design, Computer Networks, Linear Algebra. **GPA 3.88/4.00**

Experience

Assistant Research Software Engineer, jsPsych Poughkeepsie, NY | May 2025 - May 2026

- Merged upstream fixes and plugins into the open-source **jsPsych** core and contrib repos
- Set the coding standards and architecture for jsPsych-timelines, the framework's first ready-to-deploy packages
- Built 7+ gold standard experiment templates in **TypeScript/HTML/CSS**, used in Heat and Cognition research
- Automated the deployment pipeline (**Prolific** to **OSF**), cutting study turnaround from weeks to hours

Service Desk Assistant, Computing and Information Services, Vassar College Poughkeepsie, NY | Sep 2024 - Sep 2026

- Resolved **20+ tickets a week** by phone and walk-up: OS debugging and hardware repair on Mac, PC, iOS, Android

Projects

farhat: Student Guidance Platform Next.js, TypeScript, Prisma, Postgres, Zod, Playwright

Live scholarship and college-prep platform, built solo. Plan generation is **Zod**-validated tool-use against a per-generation allowlist, so the model structurally cannot invent a scholarship or a link. 16 **Prisma** models on **Postgres**, hand-written tsvector/GIN search migration, row-level security. 249 Vitest tests and 7 Playwright end-to-end specs.

retail_picker: Product Sourcing Agent Python, Claude API, OpenCLIP, Playwright, pytest

Six-stage resumable agent pipeline. Retrieval is an **OpenCLIP** cosine shortlist that pre-filters candidates before **LLM** adjudication. One prompt module, four named system prompts, model routing across two tiers. 90 pytest tests.

irc-td: IRC Server and Client C, Python, pytest, RFC 1459

My first C project, coming from Python. Built the protocol test infrastructure: a spec-conformance suite plus a robustness suite that drives the server with malformed clients to find states the RFC never specifies.

Placebo World: 2D Adventure Game Java, LibGDX, Android, JUnit

~7,000 lines of **Java** shipped to Android and desktop. I led the ECS to MVC restructure.

Leadership

House President, Jewett House: Led a 20-member team for 223 residents May 2025 - May 2026

Tech Chair, Vassar Coders++: Open-source mentoring for students Sep 2024 - May 2026

Webmaster, The Miscellany News: WordPress site for 1,000+ readers Feb 2025 - May 2026

Student Council President, Al Noor: Founded the first student council and clubs Dec 2022 - Jun 2024

Skills

Languages: C, C++, TypeScript/JavaScript, Python, Java, OCaml, Assembly

Testing & systems: pytest, Vitest, Playwright, RFC protocol implementation, UDP/TCP sockets, congestion control

Security & tooling: ECDH, ChaCha20, SHA-1 integrity, 2FA, Git/GitHub, CodePath Cybersecurity Honors